

Address

Mnemonics

0H

LDA 9H

1H

ADD AH

2H

ADD BH

3H

SUB CH

4H

OUT

5H

HLT

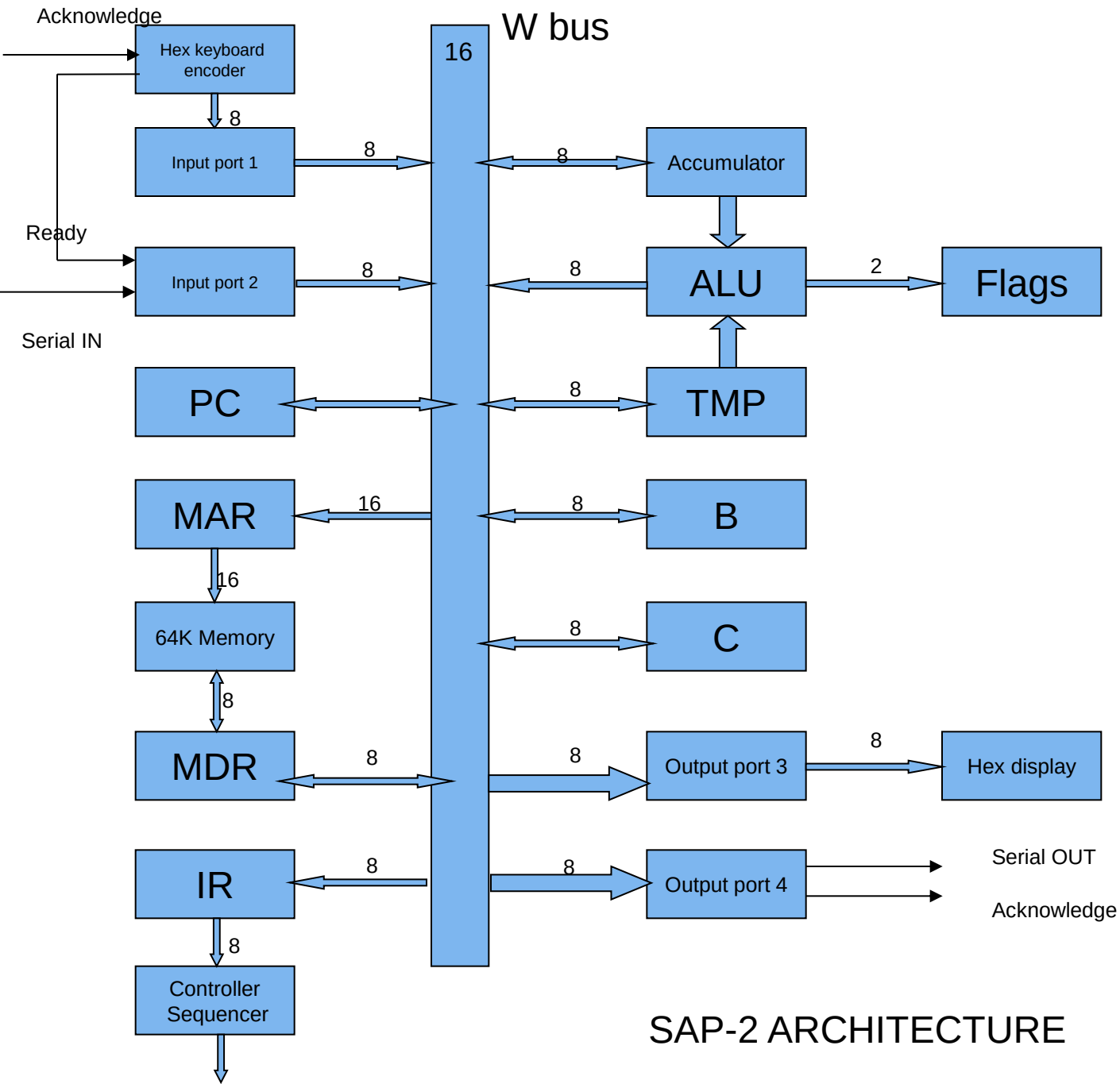
Address	Data
6H	FFH
7H	FFH
8H	FFH
9H	01H
AH	02H
BH	03H
CH	04H
DH	FFH
EH	FFH
FH	FFH

- Write an assembly language program that performs this operation:

$$8 + 4 - 3 + 5 - 2$$

Use address BH to FH for the data

SAP 2 ARCHITECTURE



SAP-2 ARCHITECTURE

Table - SAP 2 Instruction Set

Instruction	Op Code	T States	Flags	Addressing	Bytes
ADD B	80	4	S,Z	Register	1
ADD C	81	4	S,Z	Register	1
ANA B	A0	4	S,Z	Register	1
ANA C	A1	4	S,Z	Register	1
ANI byte	E6	7	S,Z	Immediate	2
CALL address	CD	18	None	Immediate	3
CMA	2F	4	None	Implied	1
DCR A	3D	4	S,Z	Register	1
DCR B	05	4	S,Z	Register	1
DCR C	0D	4	S,Z	Register	1

Table - SAP 2 Instruction Set

Instruction	Op Code	T States	Flags	Addressing	Bytes
HLT	76	5	None	None	1
IN byte	DB	10	None	Direct	2
INR A	3C	4	S,Z	Register	1
INR B	04	4	S,Z	Register	1
INR C	0C	4	S,Z	Register	1
JM address	FA	10/7	None	Immediate	3
JMP address	C3	10	None	Immediate	3
JNZ address	C2	10/7	None	Immediate	3
JZ address	CA	10/7	None	Immediate	3
LDA address	3A	13	None	Direct	3

Table - SAP 2 Instruction Set

Instruction	Op Code	T States	Flags	Addressing	Bytes
MOV A,B	78	4	None	Register	1
MOV A,C	79	4	None	Register	1
MOV B,A	47	4	None	Register	1
MOV B,C	41	4	None	Register	1
MOV C,A	4F	4	None	Register	1
MOV C,B	48	4	None	Register	1
MVI A, byte	3E	7	None	Immediate	2
MVI B, byte	06	7	None	Immediate	2
MVI C, byte	0E	7	None	Immediate	2
NOP	00	4	None	None	1

Table - SAP 2 Instruction Set

Instruction	Op Code	T States	Flags	Addressing	Bytes
ORA B	B0	4	S,Z	Register	1
ORA C	B1	4	S,Z	Register	1
ORI byte	F6	7	S,Z	Immediate	2
OUT byte	D3	10	None	Direct	2
RAL	17	4	None	Implied	1
RAR	1F	4	None	Implied	1
RET	C9	10	None	Implied	1
STA address	32	13	None	Direct	3
SUB B	90	4	S,Z	Register	1
SUB C	91	4	S,Z	Register	1

Instruction	Op Code	T States	Flags	Addressing	Bytes
XRA B	A8	4	S,Z	Register	1
XRA C	A9	4	S,Z	Register	1
XRI byte	EE	7	S,Z	Immediate	2

Seatwork

- Show the mnemonics for a program that loads the accumulator with 49H, the B register with 4AH, and the C register with 4BH; then have the program store the accumulator data at memory location 6285H and translate the foregoing program into 8080/8085 machine language, starting at address 2000H.

Seatwork

- Show the mnemonics for adding decimal 23 (17H) and 45(2D). The answer is to be stored in the C register.

Address	Symbolic	Contents
1000H	MVI A, 17H	3E
1001H		17
1002H	MVI B, 2DH	06
1003H		2D
1004H	ADD B	80
1005H	MOV C,A	4F
1006H	HLT	76

- SAP 2 has a clock frequency of 1Mhz. This means that each T state has a duration of 1 μ s. How long does it take to execute the following subroutines?